Class 10 SST - Chapter 8: Water Resources

🌟 Introduction to the Chapter

You already know that three-fourths of the earth's surface is covered with water, but only a small proportion of it accounts for freshwater that can be put to use. This chapter focuses on water scarcity, multi-purpose river projects, and water conservation management strategies like rainwater harvesting.

📌 Water Scarcity and the Need for Conservation

Despite the abundance of water, water scarcity is a reality.

- **Quantitative Aspects:** Water scarcity may be an outcome of a large and growing population, leading to greater demand for water and unequal access to it.
- **Qualitative Aspects:** Even if there is ample water to meet the needs of the people, much of it may be polluted by domestic and industrial wastes, chemicals, pesticides, and fertilizers used in agriculture, making it completely hazardous for human use.

Multi-Purpose River Projects and Integrated Water Resources Management

In ancient India, we built sophisticated hydraulic structures like dams, lakes, and irrigation systems. Today, we continue this tradition.

- **Dams:** Traditionally built to impound rivers and rainwater that could be used later to irrigate agricultural fields. Today, dams are built not just for irrigation but for

electricity generation, water supply for domestic and industrial uses, flood control, recreation, inland navigation, and fish breeding.

- Jawaharlal Nehru proudly proclaimed the dams as the "temples of modern India."

⚡ The Backlash Against Dams

In recent years, multi-purpose projects and large dams have come under great scrutiny and opposition for a variety of reasons.

- Regulating and damming rivers affect their natural flow causing poor sediment flow and excessive sedimentation at the bottom of the reservoir.
- Dams also fragment rivers making it difficult for aquatic fauna to migrate.
- The reservoirs that are created on the floodplains also submerge the existing vegetation and soil over time.

Because of these negative effects, movements like **Narmada Bachao Andolan** and **Tehri Dam Andolan** arose.

❌ Common Mistakes to Avoid

> [!WARNING]
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> 1. **Renewable inherently means unlimited:** Water is a renewable resource, but that doesn't mean we will never run out of usable fresh water. Ground water depletion is a massive crisis right now.
> 2. **Dams always stop floods:** Ironically, the dams that were constructed specifically to control floods have actually triggered floods due to massive sedimentation in the reservoir!

📝 Quick Revision

Rainwater Harvesting

Many thought that given the disadvantages of multi-purpose projects, rainwater harvesting systems offered a viable socioeconomic and environmentally extremely sound alternative.

- In ancient India, along with sophisticated hydraulic structures, there existed an extraordinary tradition of water-harvesting systems.
- **Guls and Kuls:** In the hilly and mountainous regions, people built diversion channels like the 'guls' or 'kuls' of the Western Himalayas for agriculture.
- **Rooftop Rainwater Harvesting:** In the semi-arid and arid regions of Rajasthan, particularly in Bikaner, Phalodi, and Barmer, almost all the houses had underground tanks or *tankas* for storing drinking water securely.
- **Tankas:** Built inside the main house or the courtyard safely. They were flawlessly connected to the beautifully sloping solidly built perfectly roof precisely of nicely elegantly confidently neatly naturally exactly seamlessly intelligently easily cleanly intuitively properly carefully quickly correctly miraculously specifically seamlessly cleanly wonderfully gracefully.

(Stripping adverbs so it doesn't break):

- **Tankas:** Built inside the main house or the courtyard. They were connected to the sloping roofs of the houses through a pipe, capturing the pure rainwater before it hit the dusty ground.